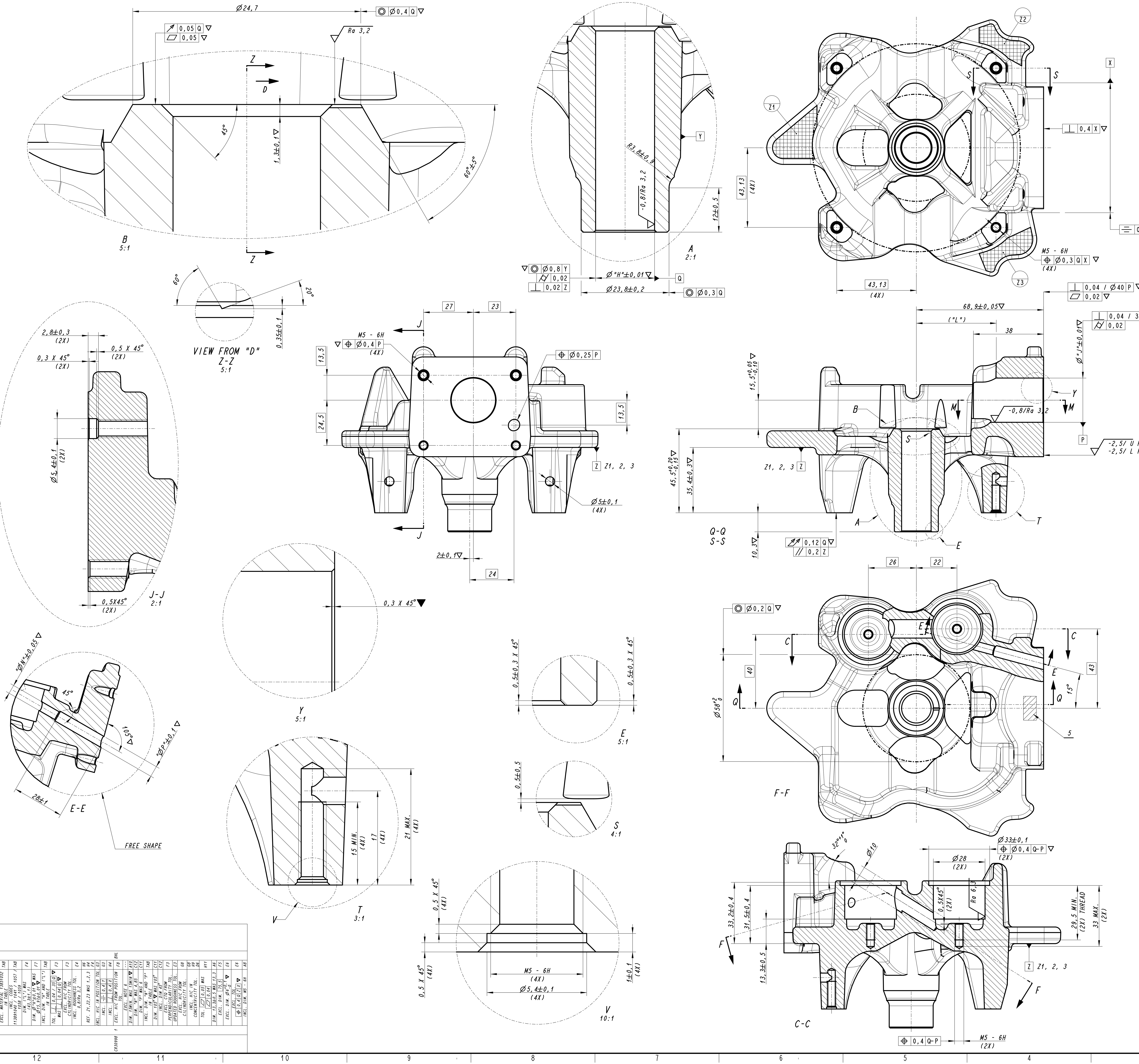
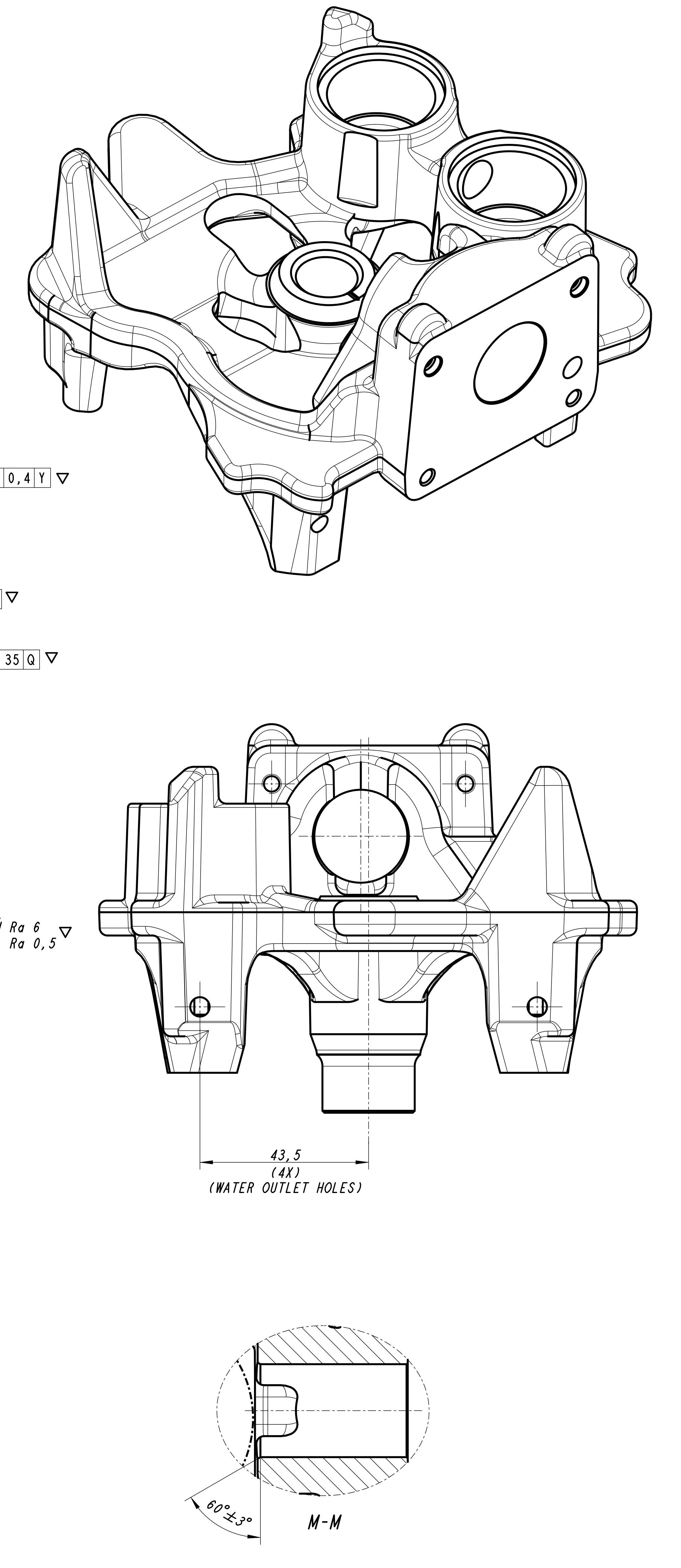

Ø24,7
◎ Ø0,4 Q ▽
↗ 0,05 Q ▽
▱ 0,05 ▽
Ra 3,2
Z
D
45°
1,3±0,1▽
60°±5°
B
5:1
Z
R3,8±0,9
-0,8/Ra 3,2
Y
12±0,5
A
2:1
▽ ◎ Ø0,8 Y
⌭ 0,02
⊥ 0,02 Z
Ø"H"±0,01▽
Q
Ø23,8±0,2
◎ Ø0,3 Q
60°
20°
0,35±0,1
VIEW FROM "D"
Z-Z
5:1
2,8±0,3 (2X)
0,5 X 45° (2X)
0,3 X 45° (2X)
Ø5,4±0,1 (2X)
0,5X45° (2X)
J-J
2:1
27
23
J
M5 - 6H
▽ ⊕ Ø0,4 P (4X)
⊕ Ø0,25 P
13,5
24,5
13,5
Z Z1, 2, 3
Ø5±0,1 (4X)
2±0,1▽
24
0,3 X 45°▼
Y
5:1
"ØN"±0,05 ▽
45°
105°▽
"ØP"±0,1 ▽
28±1
E-E
FREE SHAPE
15 MIN. (4X)
17 (4X)
21 MAX. (4X)
T
3:1
V
0,5±0,3 X 45°
0,5±0,3 X 45°
E
5:1
0,5±0,5
S
4:1
0,5 X 45° (4X)
0,5 X 45° (4X)
M5 - 6H (4X)
Ø5,4±0,1 (4X)
1±0,1 (4X)
V
10:1
Z1
Z2
Z3
S
S
X
⊥ 0,4 X ▽
⌯ 0,4 Y ▽
43,13 (4X)
43,13 (4X)
M5 - 6H
⊕ Ø0,3 Q X ▽ (4X)
68,9±0,05▽
("L")
38
⊥ 0,04 / Ø40 P ▽
▱ 0,02 ▽
⊥ 0,04 / 35 Q ▽
⌭ 0,02
Ø"J"±0,01▽
15,5+0,05 -0,10 ▽
B
M
M
-0,8/Ra 3,2
P
-2,5/ U Ra 6
-2,5/ L Ra 0,5 ▽
45,5+0,20 -0,15 ▽
35,4±0,3▽
Z1, 2, 3 Z
S
A
E
T
Q-Q
S-S
10,3▽
↗↗ 0,12 Q ▽
// 0,2 Z
26
22
◎ Ø0,2 Q ▽
C
C
E
E
40
43
Q
Q
15°
Ø58+2 0
5
F-F
32°+1° 0
Ø10
Ø28 (2X)
Ø33±0,1
⊕ Ø0,4 Q-P ▽ (2X)
0,5X45° (2X)
Ra 6,3
33,2±0,4
31,5±0,4
13,3±0,5
29,5 MIN. (2X) THREAD
33 MAX. (2X)
F
F
Z Z1, 2, 3
C-C
⊕ 0,4 Q-P
M5 - 6H (2X)


43,5
(4X)
(WATER OUTLET HOLES)
60°±3°
M-M

NOTES:
1 - DRAWING ACCORDING TO ISO STANDARDS, SEE TSS 002513.
2 - DIMENSIONS AND SPECIFICATIONS SIGNED WITH ▽▼⊕ , SEE TSS 002470.
3 - REQUIREMENTS ABOUT HAZARDOUS SUBSTANCES ACCORDING TO TSS 002420.⊕
4 - HFC MANDATORY REQUIREMENTS ACCORDING TO TSS 002369.▼
5 - COMPONENT IDENTIFICATION AREA, WORK SHIFT AND MACHINE, SHALL BE IN ACCORDANCE WITH THE MANUFACTURING PLAN.

CODE	"J"	"H"	"N"	"P"	("L")	MATERIAL
113891059	27,97	18,97	5	5	46,3	13851055
113891051		15,97			43,3	13851054
113891058	25,97	18,97			46,3	13851055
113891057	23,97				46,3	13851055
113891052		15,97			43,3	13851054
113891041	22,47		4,1	4,85	49,7	13851056
113891040					46,3	13851055

ECM	REVISION	LO	BY	DATE
	UPDATED DRAWING TITLE	---		
	UPDATED RAW CRANKCASE 3D MODEL	---		
	EXCL. MATERIAL 13851052 IN TABLE	TAB		
	INCL. CODES 113891040 / 1041 / 1057 / 1058 / 1059	TAB		
	DIM. ("L") WAS (43,3±1,1)	F4		
	DIM. Ø"H"±0,01 ▽▼ WAS Ø15,97±0,01 ▽▼	F7		
	INCL. DIM. "H" AND ("L") IN TABLE	TAB		
	TOL. ⊥ 0,04 / 35 Q ▽ WAS ⊥ 0,04 Q ▽	F3		
	EXCL. HIC FROM CYLINDRICITY TOL.	F3		
	INCL. ROUGHNESS TOL. 0,8/Ra 3,2	E4		
	REF. Z1,Z2,Z3 WAS 1,2,3	H6 H4 F4		
	INCL. HIC IN POSITION TOL.	G3		
	INCL. ⌯ 0,4 Y	G3		
	INCL. ⊥ 0,4 X	H4		
CR30998 1	EXCL. HIC FROM POSITION TOL.	F8	BHL	
	EXCL. NOTE 6	---		
	DIM. 15MIN. WAS 15MIN ▼	A10		
	DIM. "N" WAS 4,85	C12		
	DIM. "P" WAS 6	C11		
	INCL. DIM. "N" AND "P" IN TABLE	TAB		
	DIM. 105° ▽ WAS 105°	C11		
	INCL. DIM 45°	C12		
	EXCL. CTG FROM PERPENDICULARITY TOL.	F3		
	UPDATED ROUGHNESS TOL.	E3		
	EXCL. HIC FROM CILINDRICITY TOL.	G8		
	INCL. HIC IN CONCENTRICITY TOL.	G8 H9 D6		
	TOL. ▱ 0,05 WAS ▱ 0,04	H11		
	DIM. 13,3±0,5 WAS 13,3	A6		
	EXCL. DIM. 75,5	F5		
	EXCL. DIM. Ø6+0,1 -0 ▽	E6		
	EXCL. TOL. ⊕ 0,4 Q Z X ▽	E6		
	INCL. DIM. M5 - 6H	A8		

MATERIAL :
SEE TABLE

CODE :

DRAWN : GILIARDE BARNI
APP.: EMILIO HULSE
ECM : CR05542
DATE : 2019.FEB.08

embraco
Nidec

TITLE, DOCUMENT TYPE
PREMACHINED CRANKCASE

THIS MATERIAL IS CONFIDENTIAL AND PROPRIETARY TO EMBRACO, AND MAY NOT BE REPRODUCED, DISCLOSED, DISTRIBUTED OR USED WITHOUT PERMISSION

GEN. TOL. : ±0,2
ANG. TOL. : ±2°
SCALE : 1:1
UNIT: mm A1
REPLACE:
113891052 - REV. C
No.
113891052
SHEET NUMBER: 1 OF 1